

Project Proposal

Smart Campus Resource Booking System

UCS503: Software Engineering

Team Members:

Dhruv Bihani

Aditya Srivastav

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1 Introduction

Educational institutions have various resources such as classrooms, laboratories, seminar halls, projectors, and other equipment that need efficient management. Traditional manual booking methods often result in scheduling conflicts, inefficient resource utilization, and difficulties in tracking reservations.

The **Smart Campus Resource Booking System** aims to provide a centralized digital platform where students, faculty members, and administrators can efficiently manage resource availability and bookings.

This project is an extension of an existing database management system project and focuses on transforming it into a complete software application following software engineering principles.

2 Problem Statement

In many institutions, resource allocation is managed manually or through disconnected systems. This creates several issues:

- Double booking of resources.
- Lack of real-time resource availability information.
- Difficulty in managing booking requests.
- Poor tracking of resource utilization.
- Increased administrative workload.

A software solution is required to automate the booking process and provide an efficient resource management system.

3 Objectives

The main objectives of this project are:

- To develop a centralized resource booking platform.
- To allow users to view available resources and request bookings.
- To provide administrators with resource management capabilities.
- To prevent booking conflicts using automated validation.
- To maintain booking records and generate reports.

4 Proposed Solution

The proposed system will be a web-based application that provides different functionalities based on user roles.

Users will be able to:

- Register and log in.
- View available resources.
- Submit booking requests.
- Track and manage their bookings.

Administrators will be able to:

- Add, update, and remove resources.
- Approve or reject booking requests.
- Monitor resource utilization.
- Generate reports.

5 Scope of the Project

The scope of the project includes the following modules:

5.1 User Management

- User registration and authentication.
- Role-based access control.

5.2 Resource Management

- Adding and managing resources.
- Categorizing resources.
- Tracking resource availability.

5.3 Booking Management

- Checking resource availability.
- Creating booking requests.
- Approving or rejecting requests.
- Maintaining booking history.

5.4 Dashboard and Reports

- User booking dashboard.
- Admin management dashboard.
- Resource utilization reports.

6 Major Modules

6.1 Authentication Module

This module handles user registration, login, and authorization based on user roles.

6.2 Resource Module

This module manages resource details including resource name, category, and availability.

6.3 Booking Module

This module handles booking requests, approval workflow, and prevents scheduling conflicts.

6.4 Admin Module

The admin module provides functionality to manage users, resources, and bookings.

6.5 Reporting Module

This module provides booking history and resource usage statistics.

7 Technology Stack

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Java, JDBC, Servlets/JSP
- **Database:** Oracle Database
- **Version Control:** Git and GitHub
- **Documentation:** LaTeX

8 Development Methodology

The project will follow an incremental development approach.

The development phases are:

1. Requirement Analysis
2. System Design
3. Database Design
4. Prototype Development
5. Testing and Improvement
6. Final Deployment and Documentation

9 Expected Outcomes

At the end of the project, the system will provide:

- A functional web-based resource booking application.
- Efficient resource scheduling.
- Reduced booking conflicts.
- Improved resource management.
- Complete software documentation.

10 Future Enhancements

Future improvements may include:

- Email notifications.
- QR-based booking verification.
- Mobile application support.
- AI-based resource utilization prediction.

11 Team Details

Name	Role
Dhruv Bihani	Backend and Database Development
Aditya Srivastav	Frontend and Testing